

Questions

MathonGo

Q1 - 2024 (04 Apr Shift 1)

Let the length of the focal chord PQ of the parabola $y^2 = 12x$ be 15 units. If the distance of PQ from the origin is p, then $10p^2$ is equal to _____

Q2 - 2024 (04 Apr Shift 2)

Let PQ be a chord of the parabola $y^2 = 12x$ and the midpoint of PQ be at $(4, 1)$. Then, which of the following point lies on the line passing through the points P and Q?

(1) $(3, -3)$

(2) $(2, -9)$

(3) $\left(\frac{3}{2}, -16\right)$

(4) $\left(\frac{1}{2}, -20\right)$

Q3 - 2024 (05 Apr Shift 1)

Suppose AB is a focal chord of the parabola $y^2 = 12x$ of length l and slope $m < \sqrt{3}$. If the distance of the chord AB from the origin is d, then $l d^2$ is equal to _____

Q4 - 2024 (05 Apr Shift 2)

Let a line perpendicular to the line $2x - y = 10$ touch the parabola $y^2 = 4(x - 9)$ at the point P. The distance of the point P from the centre of the circle $x^2 + y^2 - 14x - 8y + 56 = 0$ is _____

Q5 - 2024 (06 Apr Shift 1)

Let a conic C pass through the point $(4, -2)$ and $P(x, y)$, $x \geq 3$, be any point on C. Let the slope of the line touching the conic C only at a single point P be half the slope of the line joining the points P and $(3, -5)$. If the focal distance of the point $(7, 1)$ on C is d, then $12d$ equals _____

Q6 - 2024 (06 Apr Shift 1)

Questions

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Let L_1, L_2 be the lines passing through the point $P(0, 1)$ and touching the parabola

$9x^2 + 12x + 18y - 14 = 0$. Let Q and R be the points on the lines L_1 and L_2 such that the $\triangle PQR$ is an isosceles triangle with base QR . If the slopes of the lines QR are m_1 and m_2 , then $16(m_1^2 + m_2^2)$ is equal to

Q7 - 2024 (09 Apr Shift 2)

Let A, B and C be three points on the parabola $y^2 = 6x$ and let the line segment AB meet the line L through C parallel to the x -axis at the point D . Let M and N respectively be the feet of the perpendiculars from A and B on L . Then $\left(\frac{AM \cdot BN}{CD}\right)^2$ is equal to _____

Q8 - 2024 (09 Apr Shift 2)

Consider the circle $C : x^2 + y^2 = 4$ and the parabola $P : y^2 = 8x$. If the set of all values of α , for which three chords of the circle C on three distinct lines passing through the point $(\alpha, 0)$ are bisected by the parabola P is the interval (p, q) , then $(2q - p)^2$ is equal to _____

Questions

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Answer Key

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Q1 (72) mathongo Q2 (4) mathongo Q3 (108) mathongo Q4 (10) mathongo

Q5 (75) mathongo Q6 (68) mathongo Q7 (36) mathongo Q8 (80) mathongo

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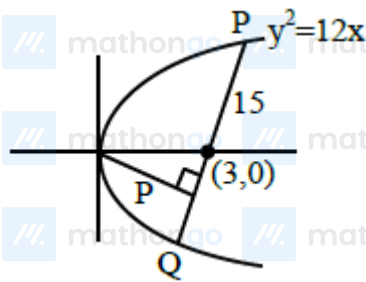
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Solutions

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Q1



$$\text{length of focal chord} = 4a \operatorname{cosec}^2 \theta = 15$$

$$12 \operatorname{cosec}^2 \theta = 15$$

$$\sin^2 \theta = \frac{4}{5}$$

$$\tan^2 \theta = 4$$

$$\tan \theta = 2$$

$$\text{equation } \frac{y-0}{x-3} = 2$$

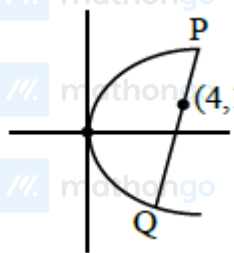
$$y = 2x - 6$$

$$2x - y - 6 = 0$$

$$P = \frac{6}{\sqrt{5}}$$

$$10p^2 = 10 \cdot \frac{36}{5} = 72$$

Q2



$$T = S_1$$

$$y - 6(x + 4)$$

$$= 1 - 48$$

$$6x - y = 23$$

Option 4 $\left(\frac{1}{2}, -20\right)$ will satisfy

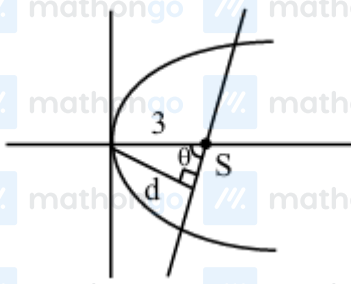
Q3

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Solutions

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$$l = 4a \operatorname{cosec}^2 \theta$$

$$l = 12 \times \frac{9}{d^2}$$

$$ld^2 = 108$$

Q4

$$y^2 = 4(x - 9)$$

$$\text{slope of tangent} = \frac{-1}{2}$$

$$\text{Point of contact } P \left(9 + \frac{1}{\left(\frac{-1}{2}\right)^2}, \frac{2 \times 1}{\frac{-1}{2}} \right)$$

$$P(13, -4)$$

$$\text{center of circle } C(7, 4)$$

$$\text{distance } CP = \sqrt{(13 - 7)^2 + (-4 - 4)^2} \\ = 10$$

Q5

$$P(x, y) \& x \geq 3$$

$$\text{Slope of line at } P(x, y) \text{ will be } \frac{dy}{dx} = \frac{1}{2} \left(\frac{y+5}{x-3} \right)$$

$$\Rightarrow 2 \frac{dy}{(y+5)} = \frac{1}{(x-3)} dx$$

$$\Rightarrow 2 \ln(y+5) = \ln(x-3) + C$$

$$\text{Passes through } (4, -2)$$

$$\Rightarrow 2 \ln(3) = \ln(1) + C$$

$$\Rightarrow C = 2 \ln(3)$$

$$\Rightarrow 2 \ln(y+5) = \ln(x-3) + 2 \ln(3)$$

$$\Rightarrow 2 \left(\ln \left(\frac{y+5}{3} \right) \right) = \ln(x-3)$$

$$\Rightarrow \left(\frac{y+5}{3} \right)^2 = (x-3)$$

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Solutions

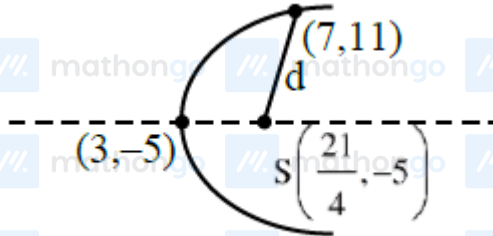
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$$\Rightarrow (y+5)^2 = 9(x-3)$$

Parabola

$$4a = 9$$

$$a = \frac{9}{4}$$



$$d = \sqrt{\left(\frac{7}{4}\right)^2 + 6^2}$$

$$d = \frac{\sqrt{625}}{4}$$

$$d = \frac{25}{4}$$

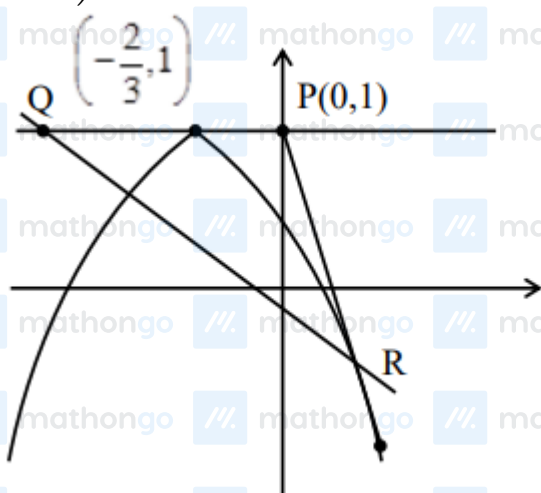
$$12d = 75$$

Q6

$$9x^2 + 12x + 4 = -18(y-1)$$

$$(3x+2)^2 = -18(y-1)$$

$$\left(x + \frac{2}{3}\right)^2 = -2(y-1)$$



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$$(0, 1)$$

$$y = mx + 1$$

$$\left(x + \frac{2}{3}\right)^2 = -2(y - 1)$$

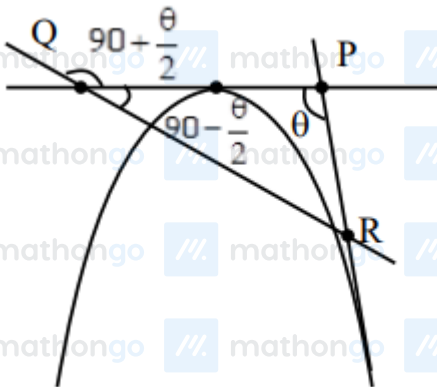
$$(3x + 2)^2 = -18mx$$

$$9x^2 + (12 + 18m)x + 4 = 0$$

$$4(6 + 9m)^2 = 4(36)$$

$$6 + 9m = 6, -6$$

$$m = 0, -\frac{4}{3}$$



$$\tan \theta = -\frac{4}{3}$$

$$\frac{2 \tan \frac{\theta}{2}}{1 - \tan^2 \frac{\theta}{2}} = \frac{-4}{3}$$

$$\left(\tan \frac{\theta}{2} - 2\right) \left(2 \tan \frac{\theta}{2} + 1\right) = 0$$

$$\tan \frac{\theta}{2} = 2, -\frac{1}{2}$$

$$m_{QR} = \tan \left(90 + \frac{\theta}{2}\right)$$

$$= -\cot \frac{\theta}{2}$$

$$m_1 = \frac{-1}{2} \quad m_2 = \frac{-1}{-1/2} = 2$$

$$16(m_1^2 + m_2^2) = 16\left(\frac{1}{4} + 4\right) = 4 + 64 = 68$$

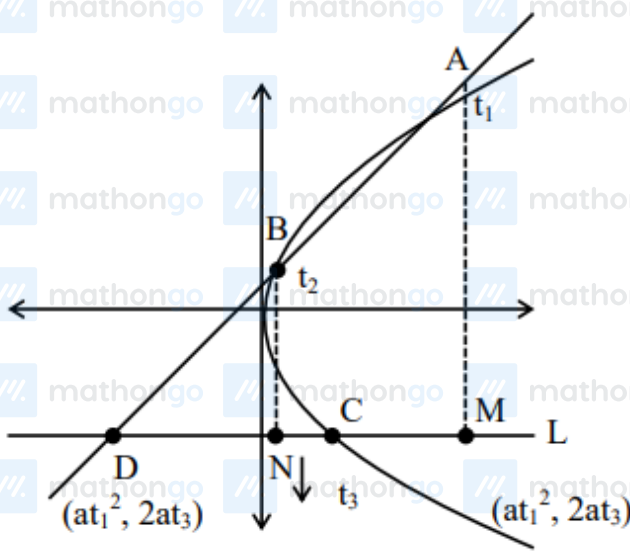
Q7

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Solutions

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$$m_{AB} = m_{AD}$$

$$\Rightarrow \frac{2}{t_1 + t_2} = \frac{2a(t_1 - t_3)}{at_1^2 - \alpha}$$

$$\Rightarrow at_1^2 - \alpha = a\{t_1^2 - t_1t_3 + t_1t_2 - t_2t_3\}$$

$$\Rightarrow \alpha = a(t_1t_3 + t_2t_3 - t_1t_2)$$

$$AM = |2a(t_1 - t_3)|, BN = |2a(t_2 - t_3)|,$$

$$CD = |at_3^2 - \alpha|$$

$$CD = |at_3^2 - a(t_1t_3 + t_2t_3 - t_1t_2)|$$

$$= a|t_3^2 - t_1t_3 - t_2t_3 + t_1t_2|$$

$$= a|t_3(t_3 - t_1) - t_2(t_3 - t_1)|$$

$$CD = a|(t_3 - t_2)(t_3 - t_1)|$$

$$\left(\frac{AM \cdot BN}{CD}\right)^2 = \left\{\frac{2a(t_1 - t_3) \cdot 2a(t_2 - t_3)}{a(t_3 - t_2)(t_3 - t_1)}\right\}^2$$

$$16a^2 = 16 \times \frac{9}{4} = 36$$

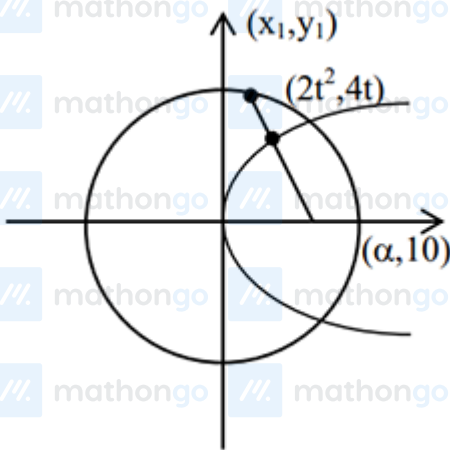
Q8

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Solutions

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$$T = S_1$$

$$x_1 + y_1 = x_1^2 + y_1^2$$

$$\alpha x_1 = x_1^2 + y_1^2$$

$$\alpha (2t^2) = 4t^4 + 16t^2$$

$$\alpha = 2t^2 + 8$$

$$\frac{\alpha - 8}{2} = t^2$$

$$\text{Also, } 4t^4 + 16t^2 - 4 < 0$$

$$t^2 = -2 + \sqrt{5}$$

$$\alpha = 4 + 2\sqrt{5}$$

$$\therefore \alpha \in (8, 4 + 2\sqrt{5})$$

$$\therefore (2q - p)^2 = 80$$

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